

TEAM ANALYTICS RACING CHALLENGE

SEASON RECAP

McQueen Racing: Team Data Analytics Racing Challenge 2026

R1

SPA

Spain

R2

NED

Netherlands

R3

USA

United States

R4

JPN

Japan

R5

GER

Germany

THE MCQUEEN RACING TEAM



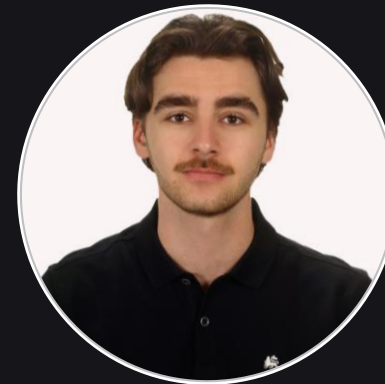
Yufei Xu

- M.Sc. Management & Tech
- Background: Quant Trading, LLM algo development



Stephen Hoffmann

- M.Sc. Consumer Science
- Background: Data Analysis and Agile Working



Thomas Rehrl

- M.Sc. Consumer Science
- Background: Data Analysis and Agile Working



Andreas Lammers

- M.Sc. Consumer Science
- Background: Business development and Retail CX

What we will cover

A Introduction & Responsibilities

B Race Results & Setup

C Car Setup Optimization Workflow

D Wrap-Up & Conclusion

Mission

- How do we get our team to perform at it's very best?
- How do we make the best possible decisions?
- How do we beat our competitors in this racing season?

01

Data Analytics

Using historical data to find the best setup for our car with iterating learning loops.

02

Strategy

Getting the most out of our car setup by maximizing the utility of fuel, tire and pit stops.

03

Team Management

Aligning communication and creating room for productive discussions.

ROLES & RESPONSIBILITIES

The functions within the team

01

Performance Analyst

Compares forecast to actual and proposes the data projects that move team performance.



02

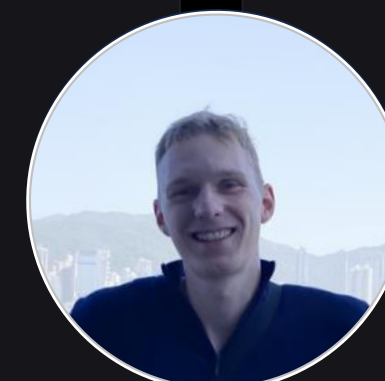
Data Principal & Systems Director

Leads the main analyses and, with the team, scopes the performance projects the analyst carries out. Curates available data so it's easy to analyze,

03

Head of Track Ops

Optimizes track operations, including refueling, tire management and pit strategy



04

Team Principal

Coordinates the team, keeps a broad overview and coordinates meetings & communication.



RESULT & SETUPS

Spain Grand Prix · 68 laps

McQueen Racing

RESULT

P4 +0

FINISHING POSITION

Avg race pace 1:28:625

Points scored 12

CHALLENGES

- No clear method approaching the setup
- Claude wasn't fully reliable in it's consulting

CAR SETUP

Rear Wing	200
Engine Mapping	200
Front Wing	200
Brake Balance	150
Differential	20
Suspension	20

STRATEGY





RESULT & SETUPS

Netherlands Grand Prix · 93 laps

McQueen Racing

RESULT

P3 +1

FINISHING POSITION

Avg race pace 1:05:858

Points scored 15

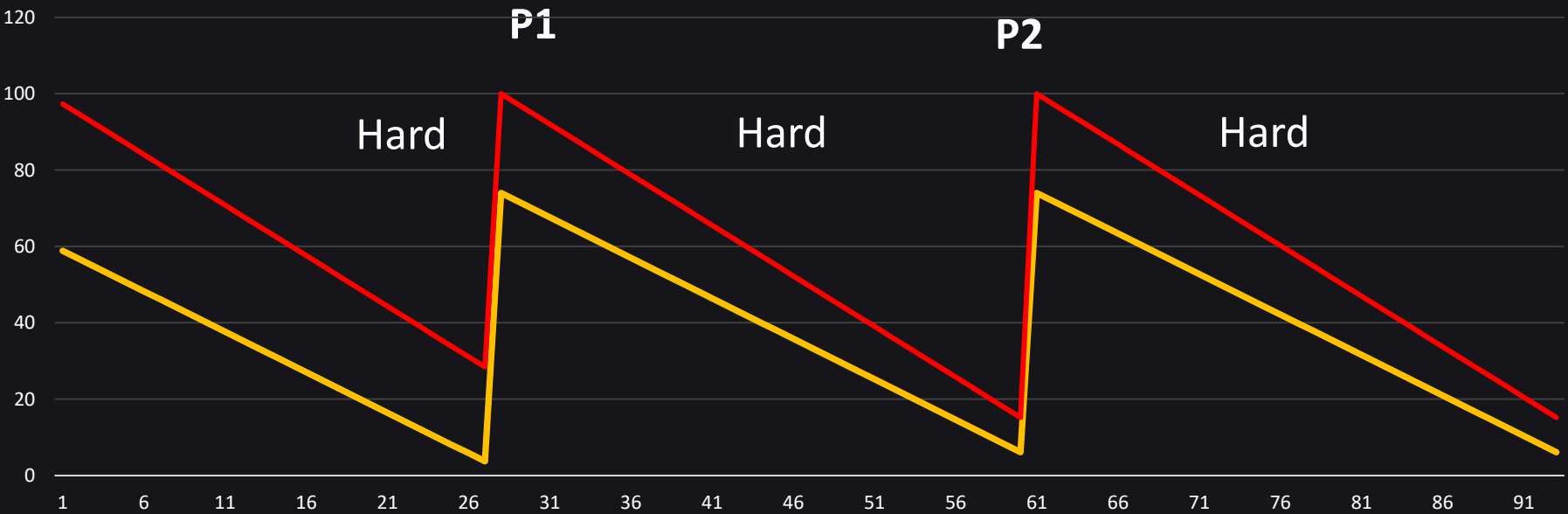
CHALLENGES

- Model was build without knowing the behavior of fuel and tire degradation

CAR SETUP

Rear Wing	489
Engine Mapping	1
Front Wing	465
Brake Balance	30
Differential	10
Suspension	40

STRATEGY



RESULT

P4-1

FINISHING POSITION

Avg race pace	1:27:970
Points scored	12

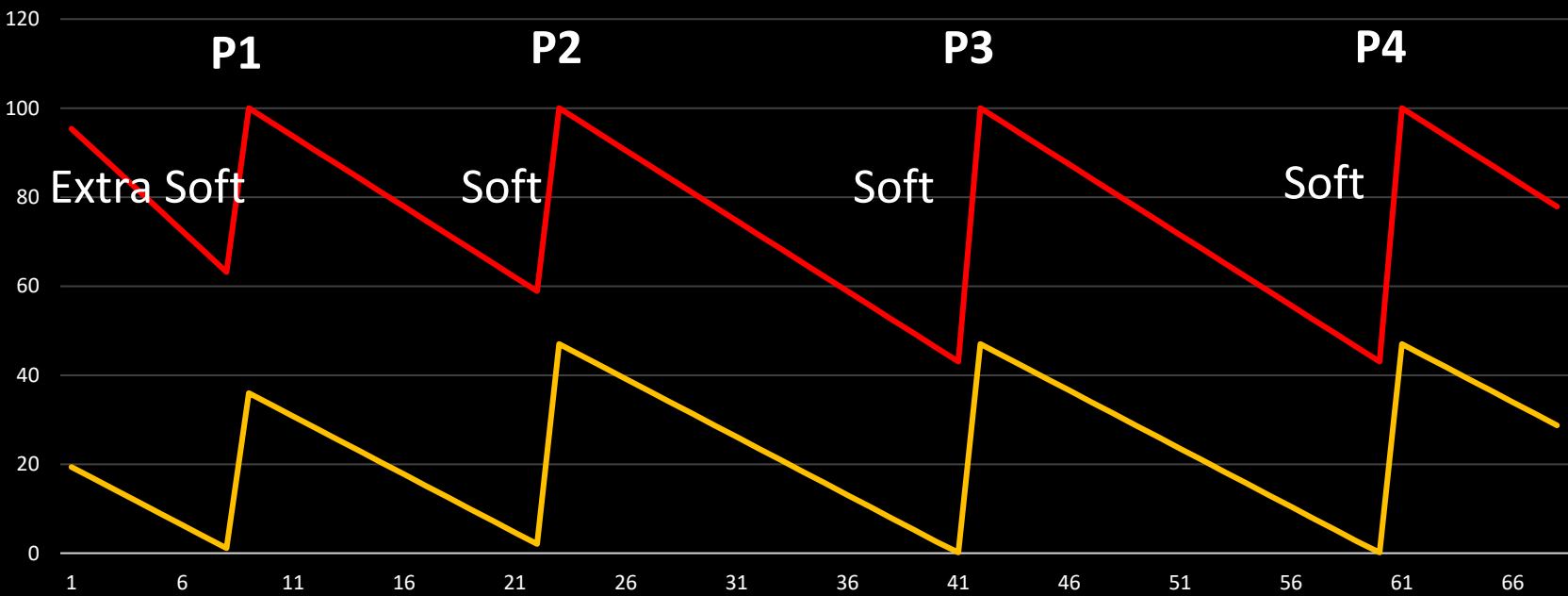
CHALLENGES

- Time constraints and no real improvement to last week
- Careless mistake in the strategy costing us the whole race

CAR SETUP

Rear Wing	237
Engine Mapping	1
Front Wing	44
Brake Balance	2
Differential	10
Suspension	152

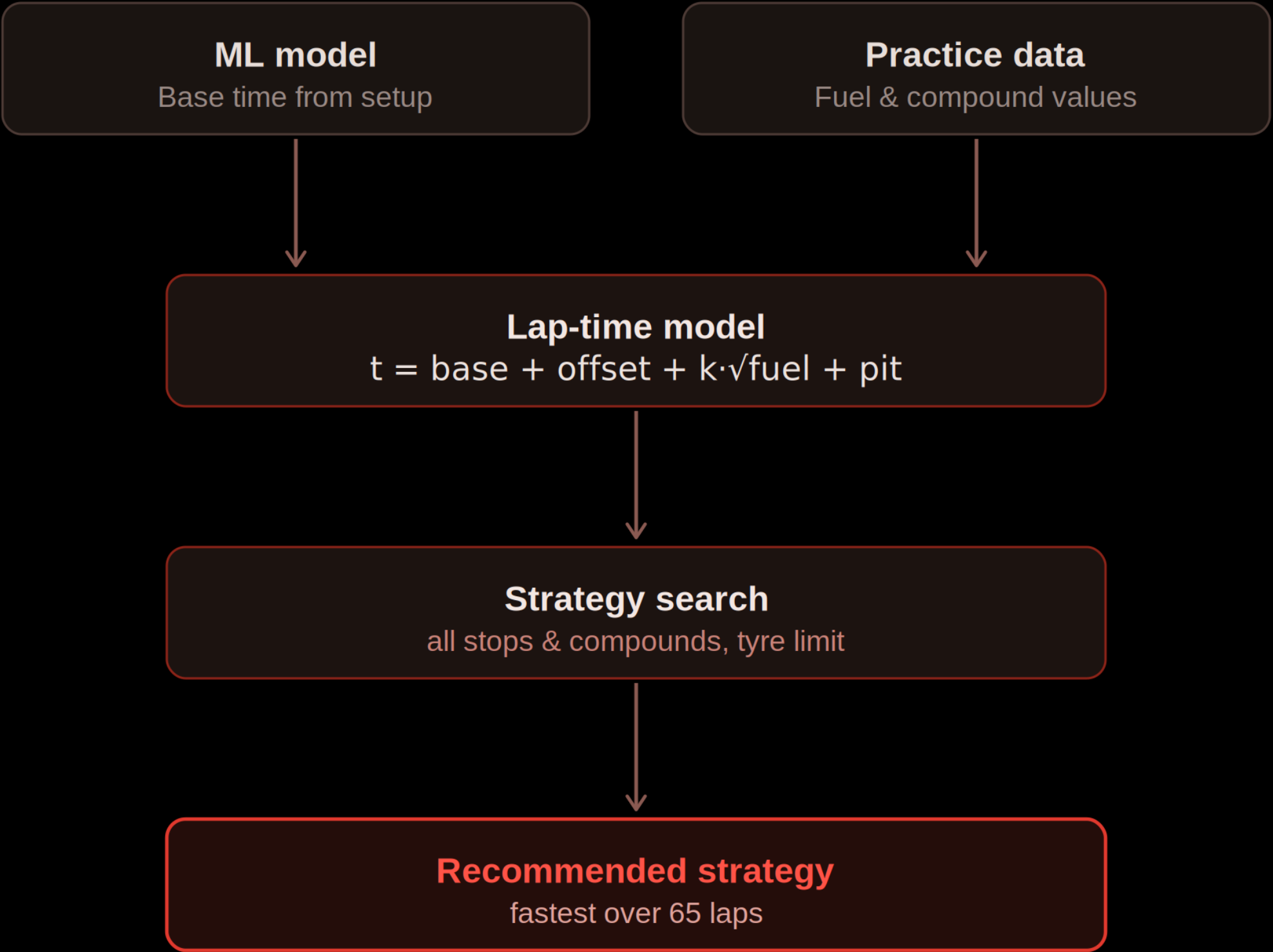
STRATEGY





STRATEGY OPTIMIZATION WORKFLOW

USA Grand Prix · 68 laps





RESULT & SETUPS

Japan Grand Prix · 65 laps

McQueen Racing

RESULT

P3 +1

FINISHING POSITION

Avg race pace 1:33:400

Points scored 15

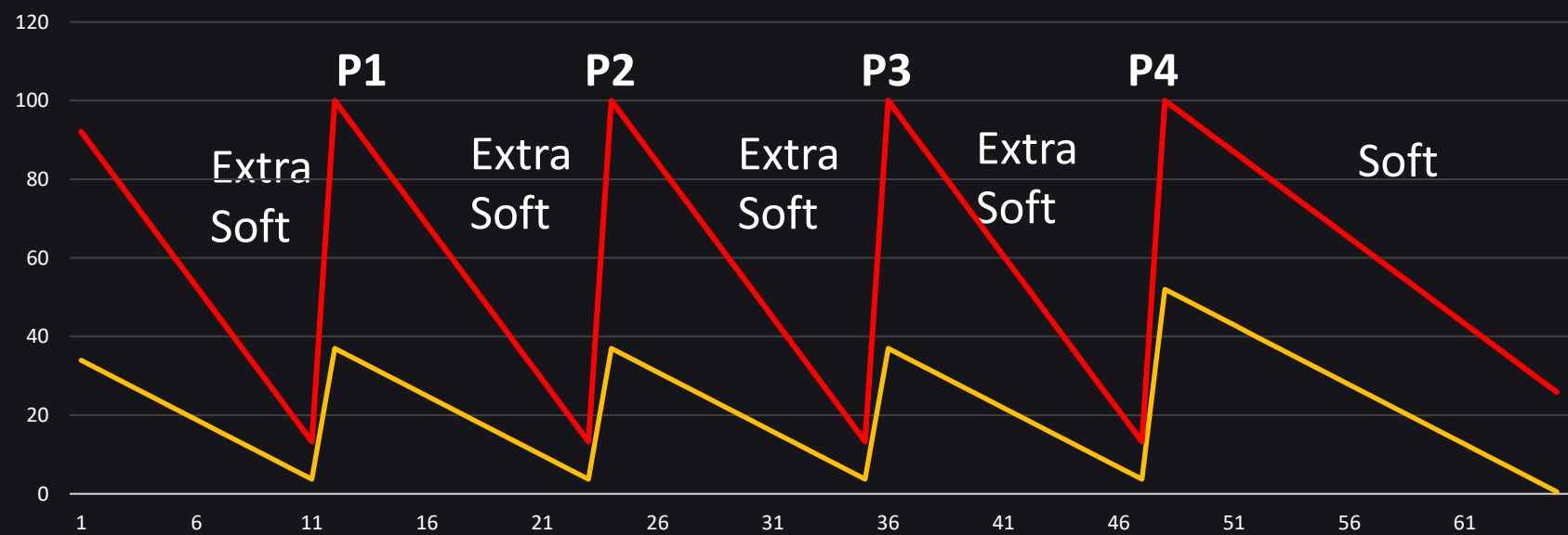
CHALLENGES

- Even though strategy now worked fine the setup was not competitive on this track → further refinement necessary.

CAR SETUP

Rear Wing	500
Engine Mapping	1
Front Wing	400
Brake Balance	75
Differential	200
Suspension	100

STRATEGY





RESULT & SETUPS

Germany / Italy · 68 laps

McQueen Racing

RESULT

P2 +1

FINISHING POSITION

Avg race pace 1:34:882

Points scored 18

CHALLENGES

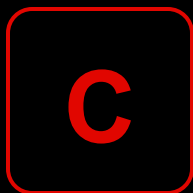
- Unclear Pitting rules

CAR SETUP

Rear Wing	1
Engine Mapping	1
Front Wing	1
Brake Balance	1
Differential	1
Suspension	1

STRATEGY

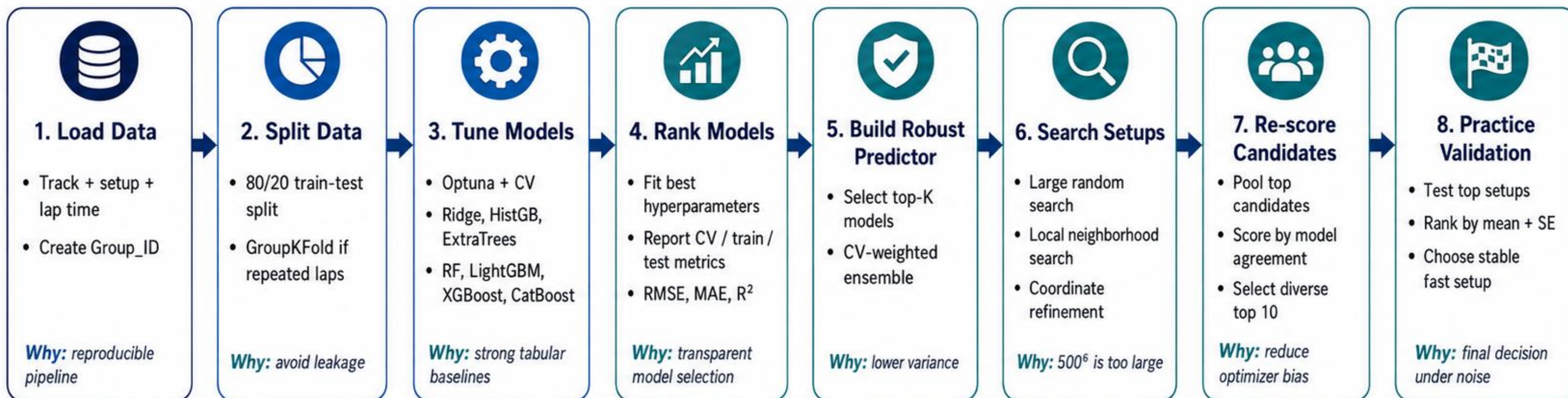




CAR SET UP OPTIMIZATION WORKFLOW

Car Setup Configuration approach

McQueen Racing



Models propose candidates; practice laps choose the final setup.



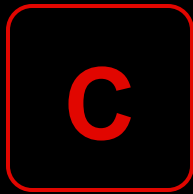
Why this works

- End-to-end methodology
- Justified model choices
- Robust to noisy data
- Clear decision logic



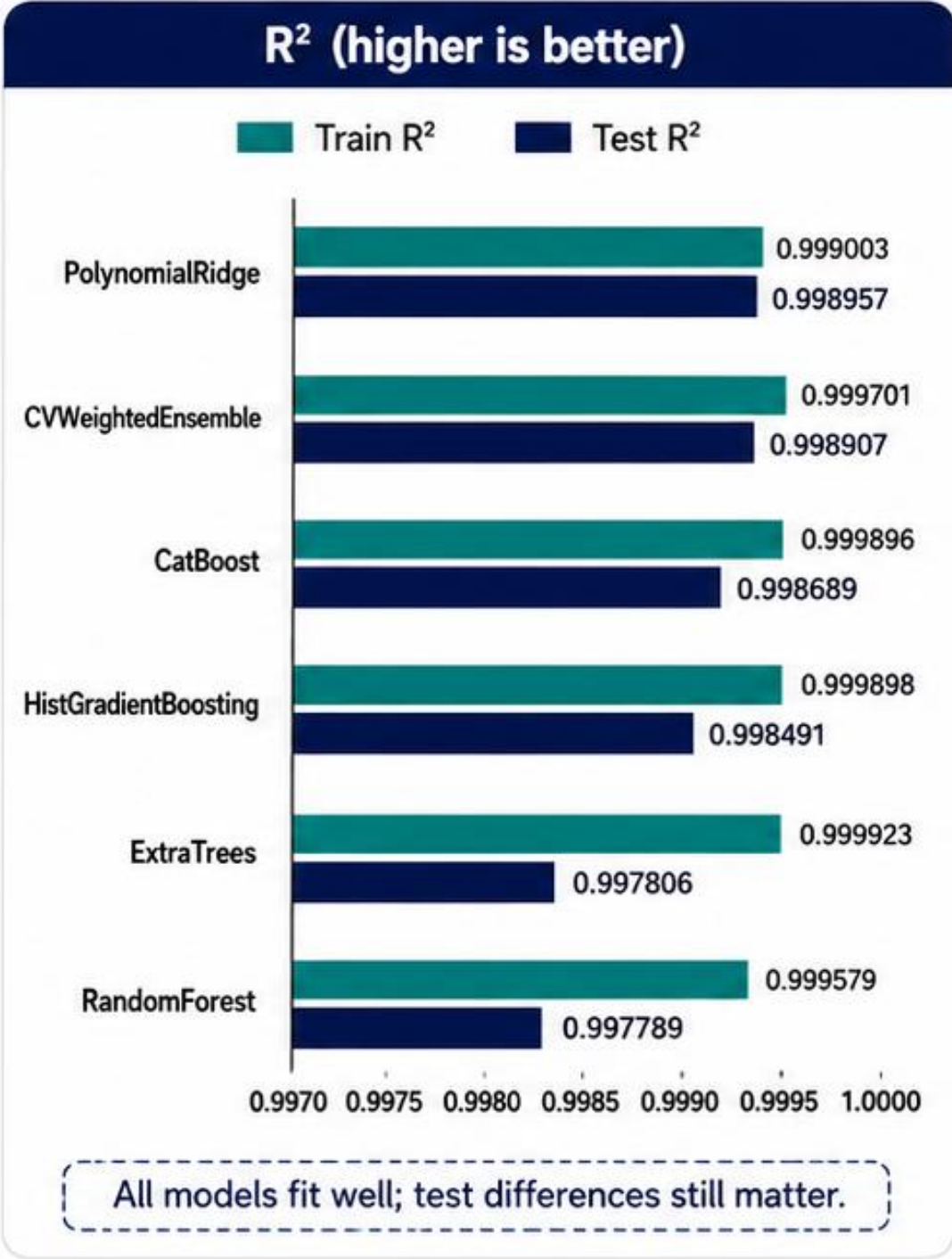
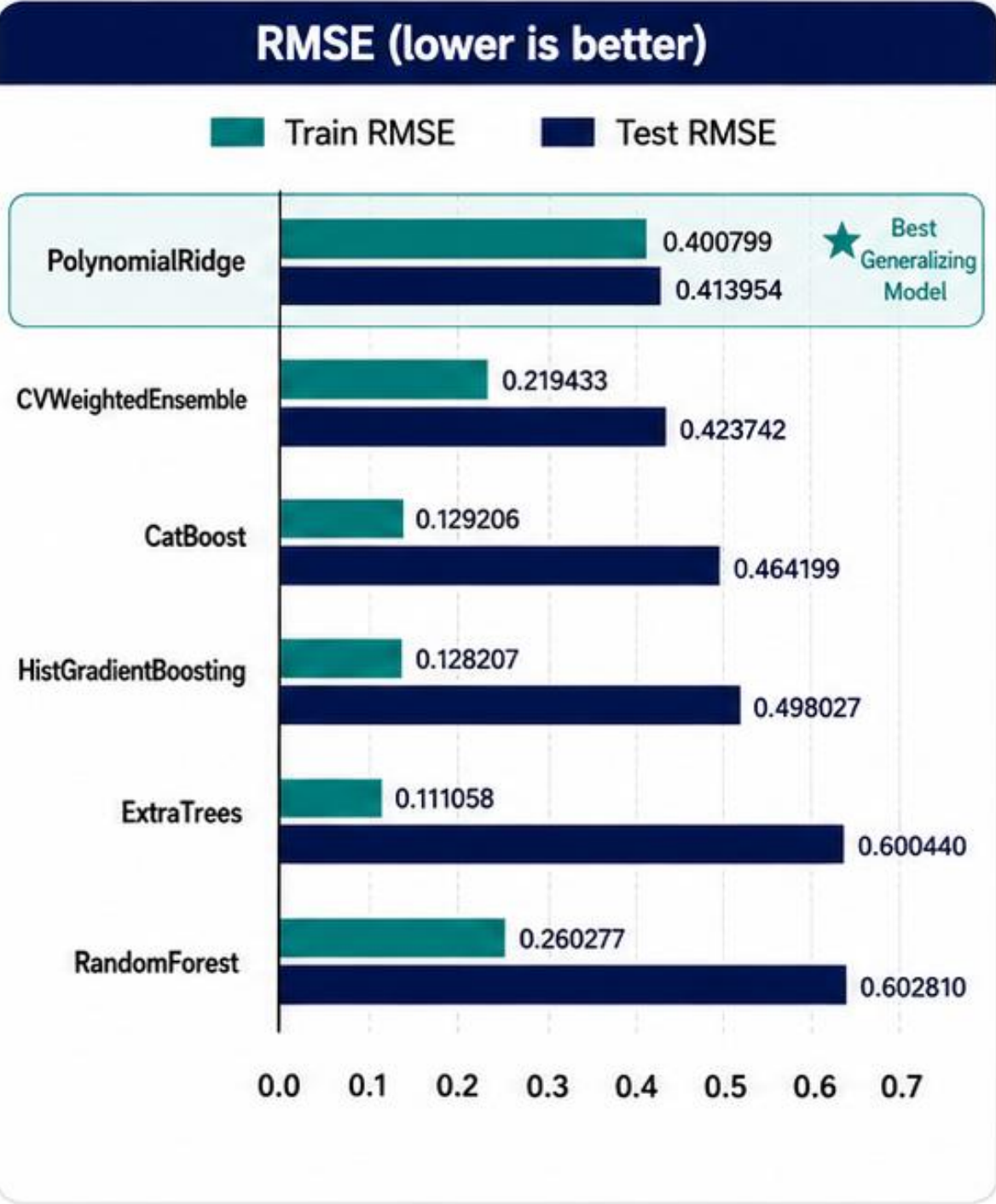
Key references

1. Breiman (1996) — *Bagging Predictors*
2. Kohavi (1995) — *Cross-validation for model selection*
3. Bergstra & Bengio (2012) — *Random search*
4. Storn & Price (1997) — *Differential evolution*



MODEL COMPARSION: Accuracy vs Generalization

Car Setup Configuration approach



Summary table				
Model	CV RMSE	Test RMSE	Test R²	Gap (Test - Train RMSE)
★ PolynomialRidge	0.416	0.414	0.998957	0.013
CVWeightedEnsemble	0.426	0.424	0.998907	0.204
CatBoost	0.470	0.464	0.998689	0.335
HistGradientBoosting	0.503	0.498	0.998491	0.370
ExtraTrees	0.603	0.600	0.997806	0.489
RandomForest	0.605	0.603	0.997789	0.343

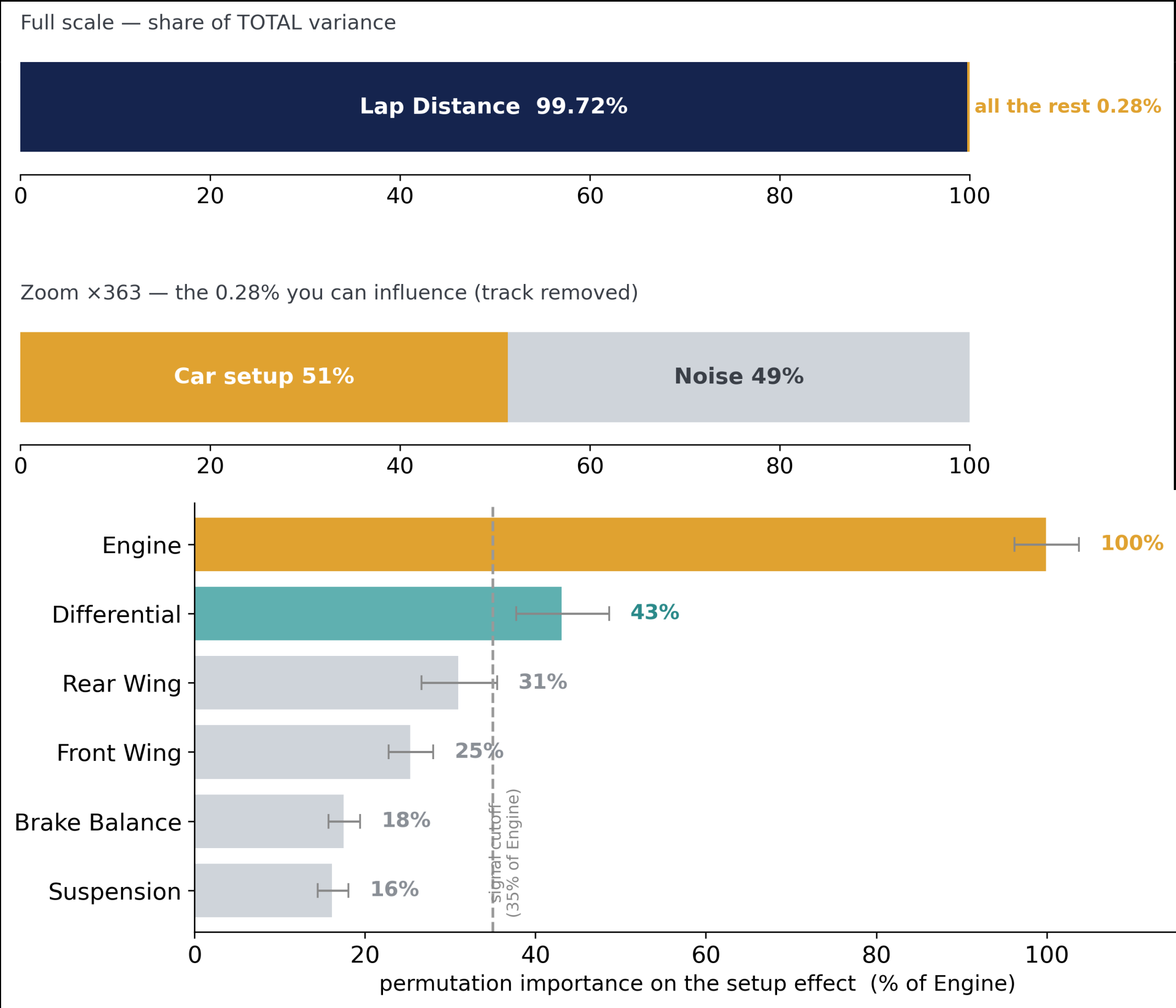
What this shows

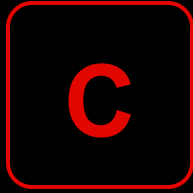
- ✓ Lowest train error is not enough
- ✓ Tree ensembles fit train data extremely well
- ✓ Several show larger test error / generalization gap
- ✓ PolynomialRidge has the best CV/Test behavior
- ✓ Model choice should prioritize CV + test performance

★ Decision logic: choose the model that generalizes best, not the one that only fits training data best.



WHERE THE LAP TIME SIGNAL LIVES

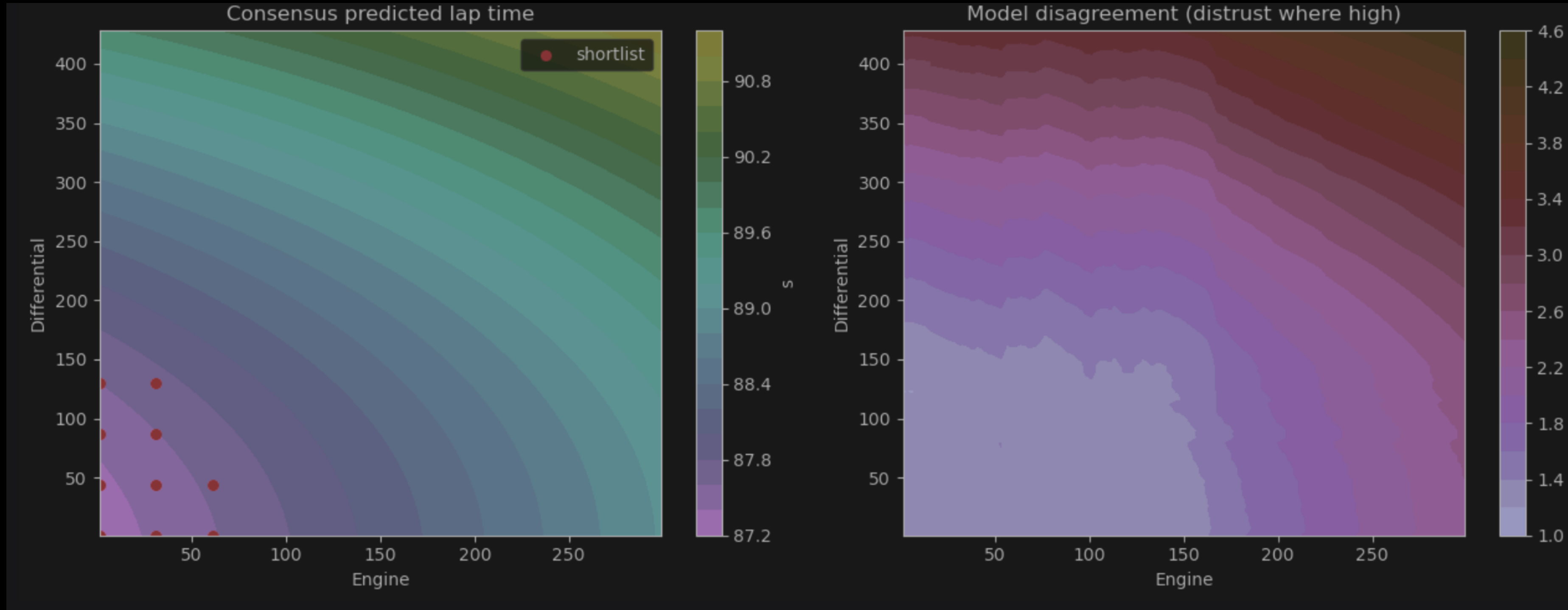


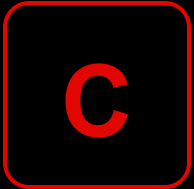


CAR SET UP OPTIMIZATION WORKFLOW

Car Setup Configuration approach

McQueen Racing





The final car

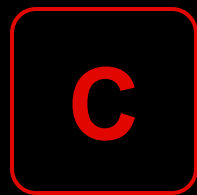


CAR SETUP

Rear Wing	1
Engine Mapping	1
Front Wing	1
Brake Balance	1
Differential	1
Suspension	1

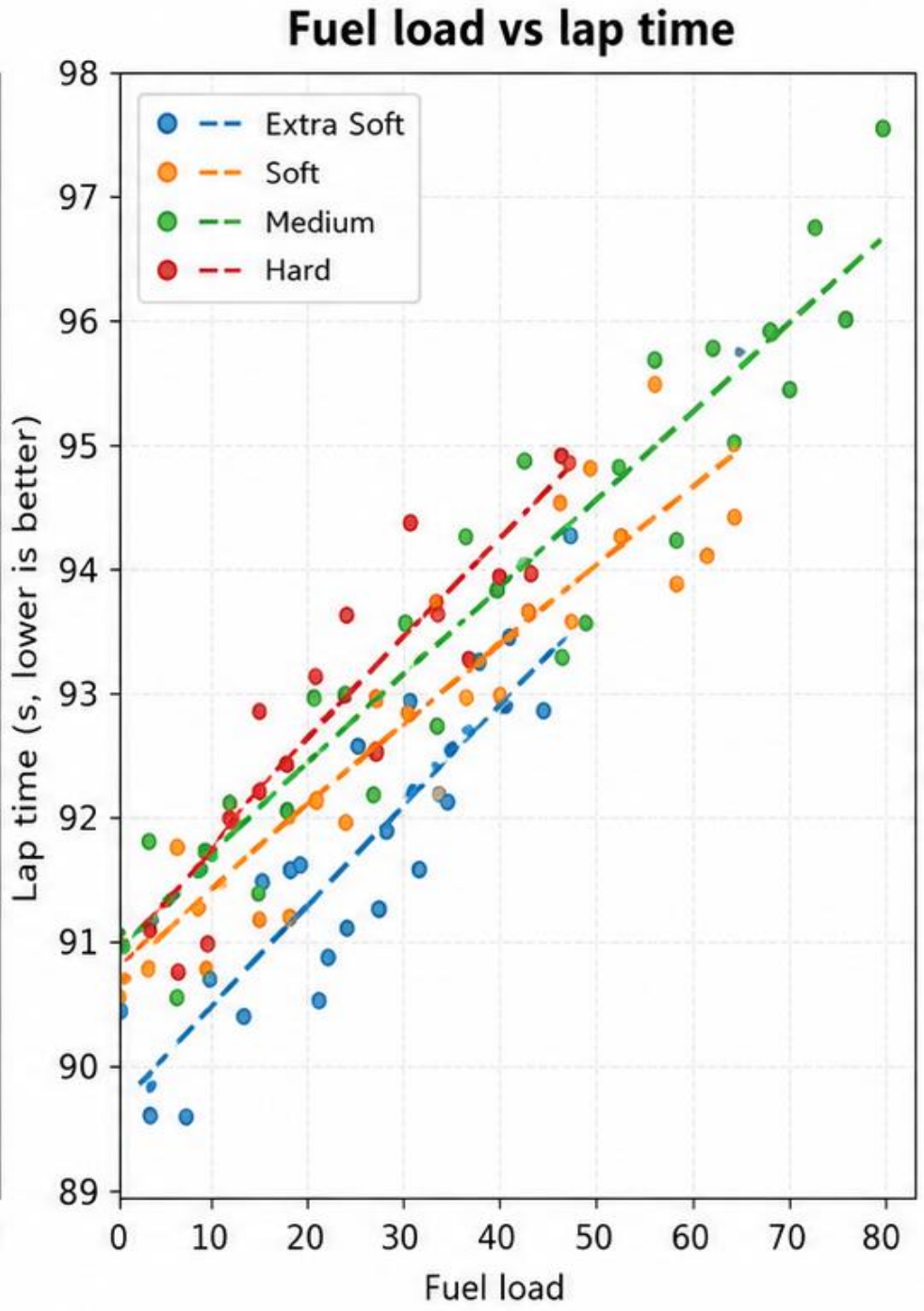
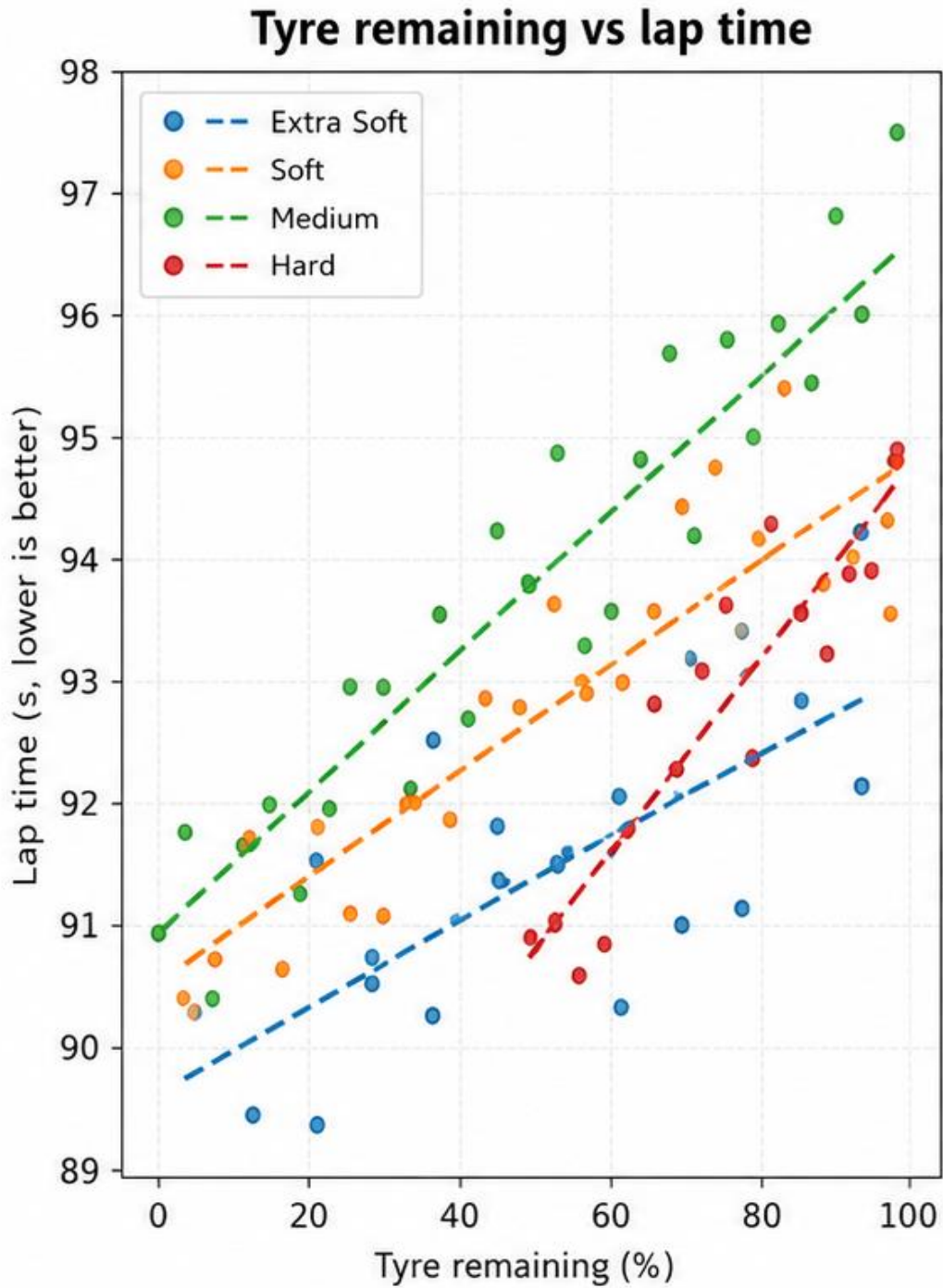
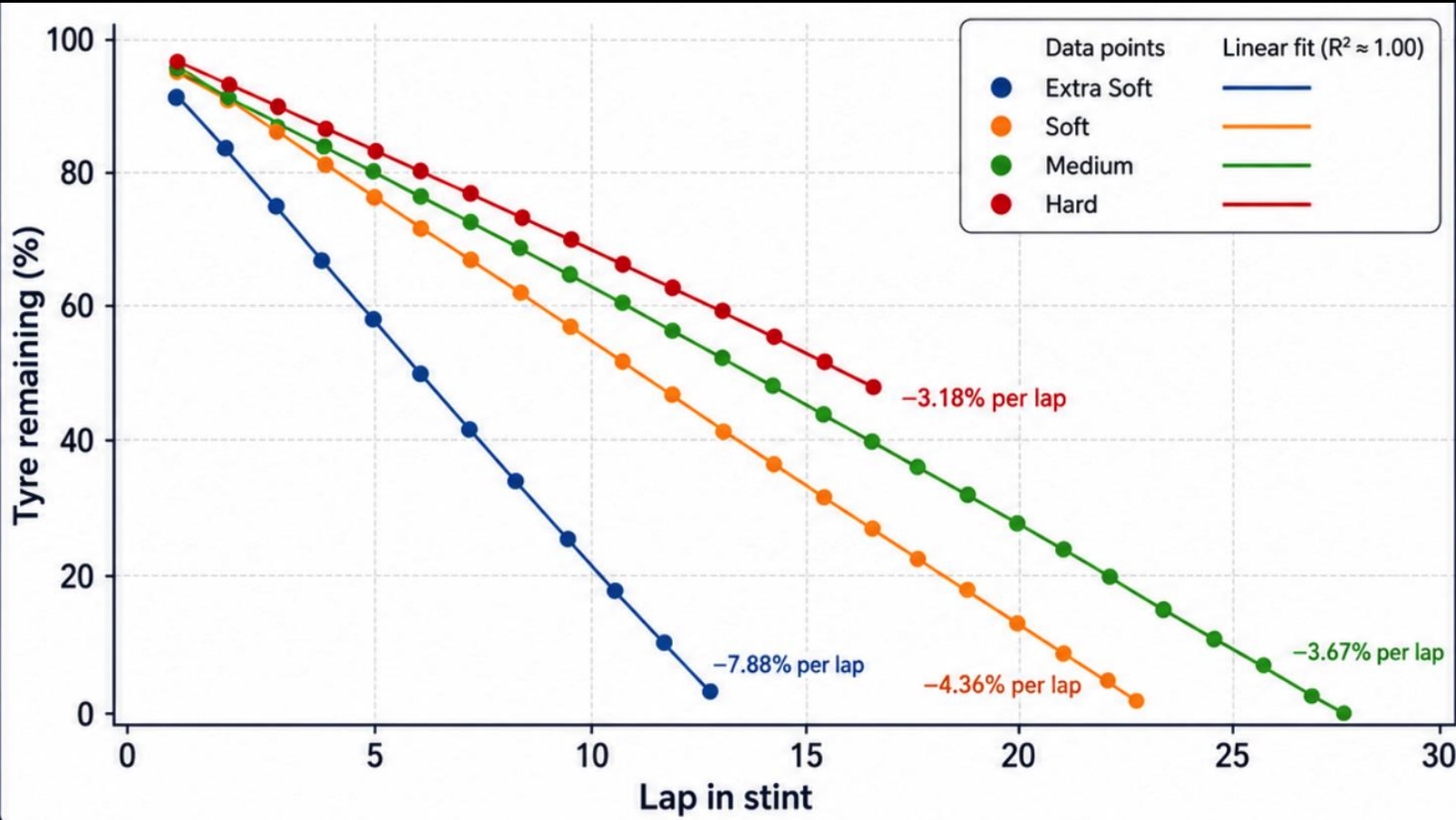
Starting Grid

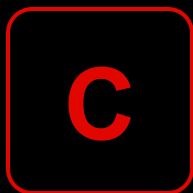
Position	Team	Lap Time	Gap
1	⚡ McQueen Racing	1:23:818	-
2	🐉 FlotterOtter	1:23:882	+0.064
3	🦅 TUMuch Pressure	1:24:188	+0.306
4	🏎️ Race To The Clouds	1:31:449	+7.261



DEGRADATION BEHAVIOR

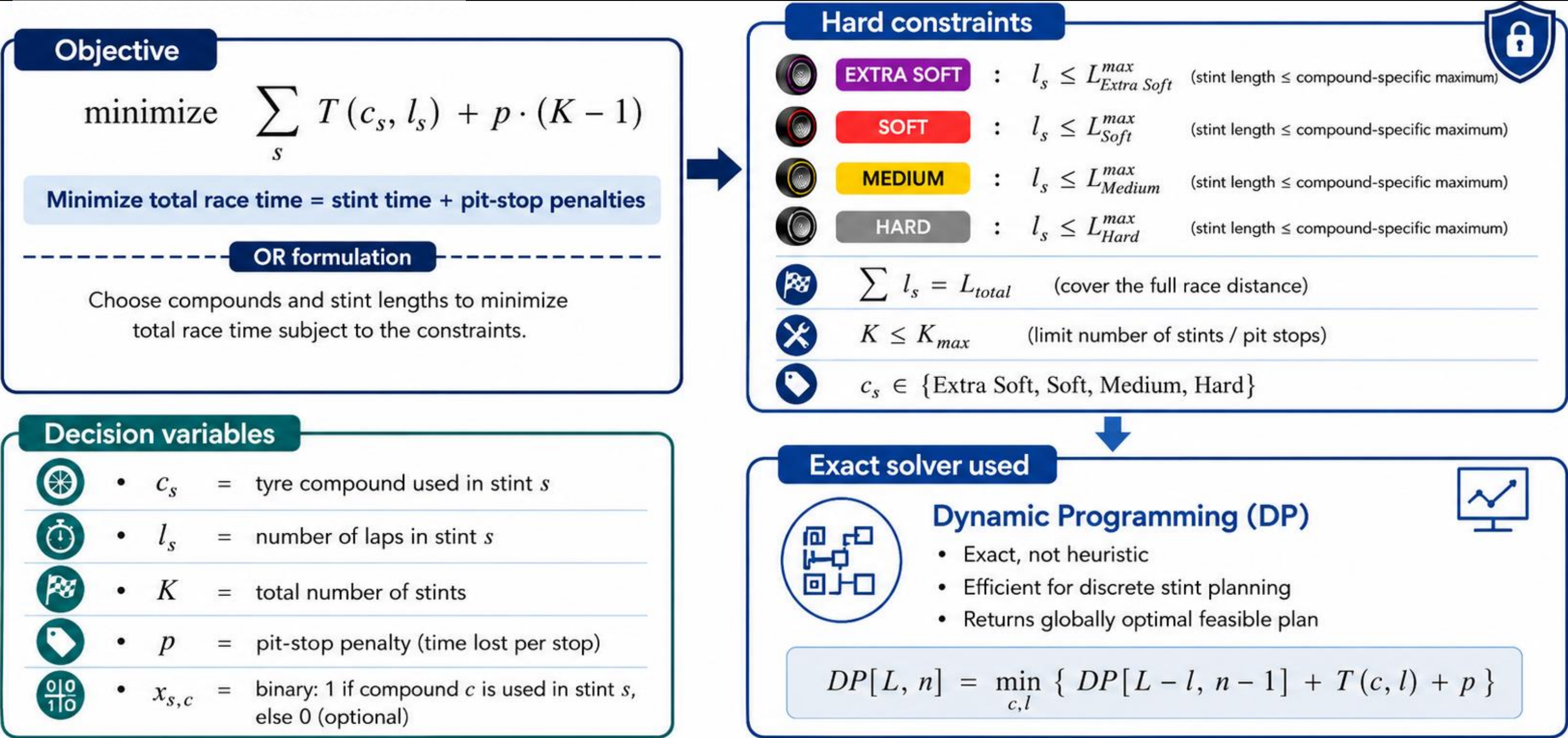
Germany / Italy · 68 laps





STRATEGY OPTIMIZATION WORKFLOW

Germany / Italy · 68 laps



WRAP UP

THEN

CRUCIBLE

Challenging initial landscape

Multi-variable problem

Fierce competition

NOW

ASCENSION

Systematic approach

Sophisticated analytics paired human insights

Team spirit

FUTURE

CAPITALIZATION

Proven repeatable methodology

Augmenting capability – combining tech & intuition

Battle tested & resilient team